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Class: DSC630 - Predictive Analytics

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Purpose: This notebook performs the tasks outlined in 3.2 Assignment: Using Data to Improve MLB Attendance. Data from Dodgers 2022 season will be used and insights will be drawn to improve attendance.

In [11]:
```
# Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import f_oneway

# Suppress warnings for cleaner output
import warnings
warnings.filterwarnings("ignore")

# Ensure that all code cells are fully visible without scrolling
from IPython.core.display import display, HTML

display(HTML('''
<style>
div.CodeMirror pre {
    white-space: pre !important;
}
</style>
'''))
```

In [12]:
```
# Read data in
df = pd.read_csv('dodgers-2022.csv')
```

In [13]:
```
# Display the first few rows of the dataframe to understand its structure
df.head()
```

Out[13]:

	month	day	attend	day_of_week	opponent	temp	skies	day_night	cap	shirt
0	APR	10	56000	Tuesday	Pirates	67	Clear	Day	NO	NO
1	APR	11	29729	Wednesday	Pirates	58	Cloudy	Night	NO	NO
2	APR	12	28328	Thursday	Pirates	57	Cloudy	Night	NO	NO
3	APR	13	31601	Friday	Padres	54	Cloudy	Night	NO	NO
4	APR	14	46549	Saturday	Padres	57	Cloudy	Night	NO	NO

Task: Get a sense of the range of attendance values and the overall distribution of attendance.

In [14]:
```
min_attend = df['attend'].min()
max_attend = df['attend'].max()

print(f"Minimum attendance: {min_attend}")
print(f"Maximum attendance: {max_attend}")
```

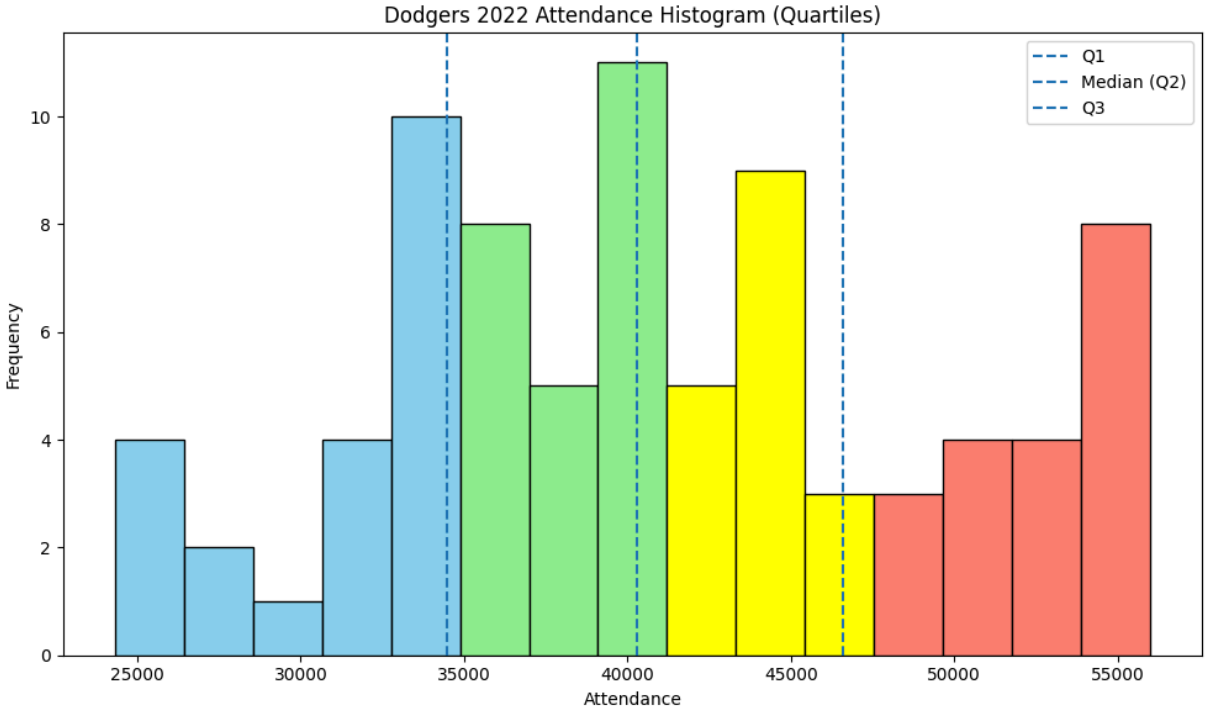
```
Minimum attendance: 24312
Maximum attendance: 56000
```

In [15]:
```
# Calculate quartiles
q1 = df['attend'].quantile(0.25)
q2 = df['attend'].quantile(0.5)
q3 = df['attend'].quantile(0.75)

# Plot histogram with quartile coloring
plt.figure(figsize=(10,6))
n, bins, patches = plt.hist(df['attend'], bins=15, edgecolor='black')

# Color bins by quartile
for patch, left_edge in zip(patches, bins[:-1]):
    if left_edge < q1:
        patch.set_facecolor('skyblue')
    elif left_edge < q2:
        patch.set_facecolor('lightgreen')
    elif left_edge < q3:
        patch.set_facecolor('yellow')
    else:
        patch.set_facecolor('salmon')

plt.axvline(q1, linestyle='--', label='Q1')
plt.axvline(q2, linestyle='--', label='Median (Q2)')
plt.axvline(q3, linestyle='--', label='Q3')
plt.xlabel('Attendance')
plt.ylabel('Frequency')
plt.title('Dodgers 2022 Attendance Histogram (Quartiles)')
plt.legend()
plt.tight_layout()
plt.show()
```



In [16]:
```
# Calculate the skewness using pandas
skewness = df['attend'].dropna().skew()
print(f"Skewness (pandas): {skewness:.3f}")
```

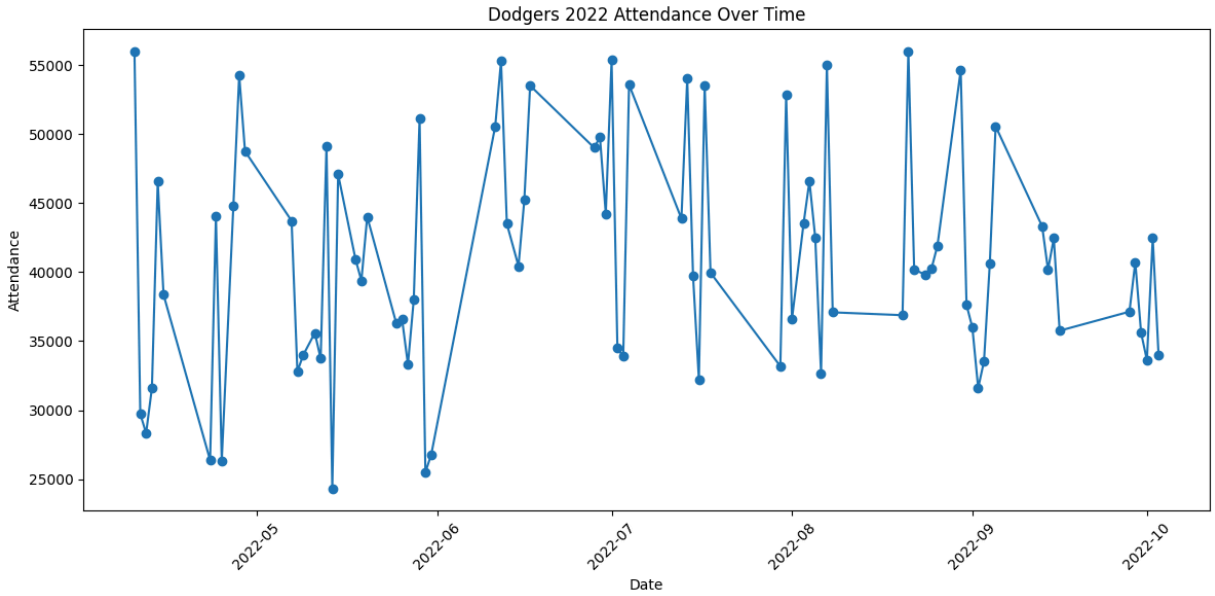
```
Skewness (pandas): 0.138
```

Conclusion: The minimum attendance is 24,312 and the maximum attendance is 56,000. The minimum attendance occurrence only happened once whereas the maximum attendance happened 8 times. The distribution is slightly skewed right.

Task: To better understand the attendance, we'll create a line chart showing the historical attendance over the season. This will also help us identify any potential trends over time.

In [17]:
```
# Create a date column from 'month' and 'day'
df['date'] = pd.to_datetime(df['month'] + ' ' + df['day'].astype(str) + ' 20

# Plot attendance over time
plt.figure(figsize=(12, 6))
plt.plot(df['date'], df['attend'], marker='o')
plt.xlabel('Date')
plt.ylabel('Attendance')
plt.title('Dodgers 2022 Attendance Over Time')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Conclusion: There appear to be maximum attendance spikes slightly above 55,000. The early season's minimum attendance is about 25,000 while the mid to end season is about 33,000. There is no clear trend over time.

Task: We will create bar charts for various categorical and binary columns to visualize the distribution of attendance based on these factors.

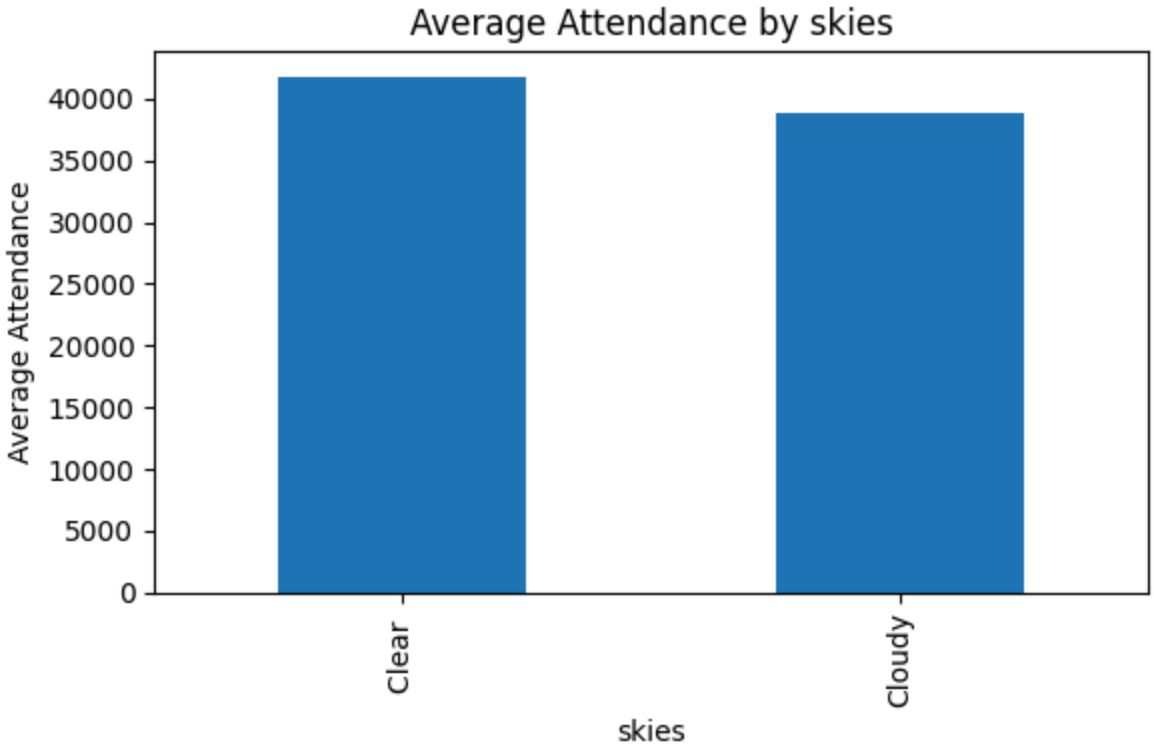
In [18]:
```
# Add a bin column for temperature because
df['temp_bin'] = pd.cut(df['temp'], bins=5)

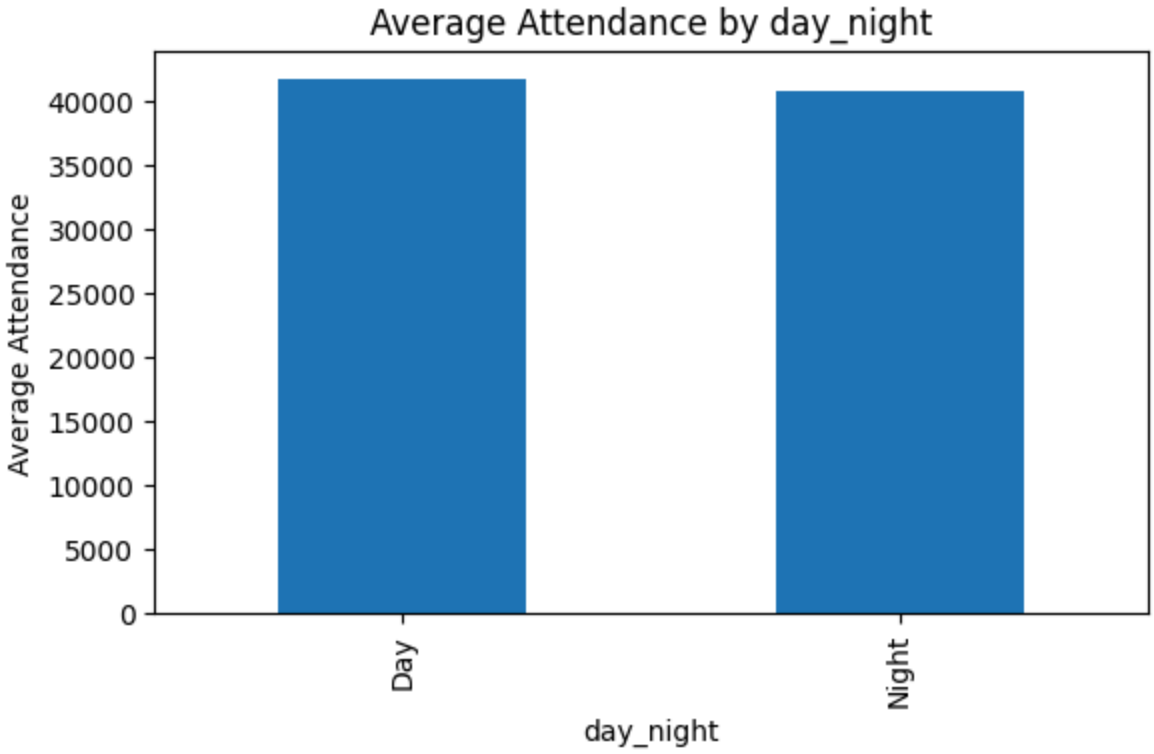
columns_to_check = ['skies', 'day_night', 'fireworks', 'cap', 'shirt', 'bobb

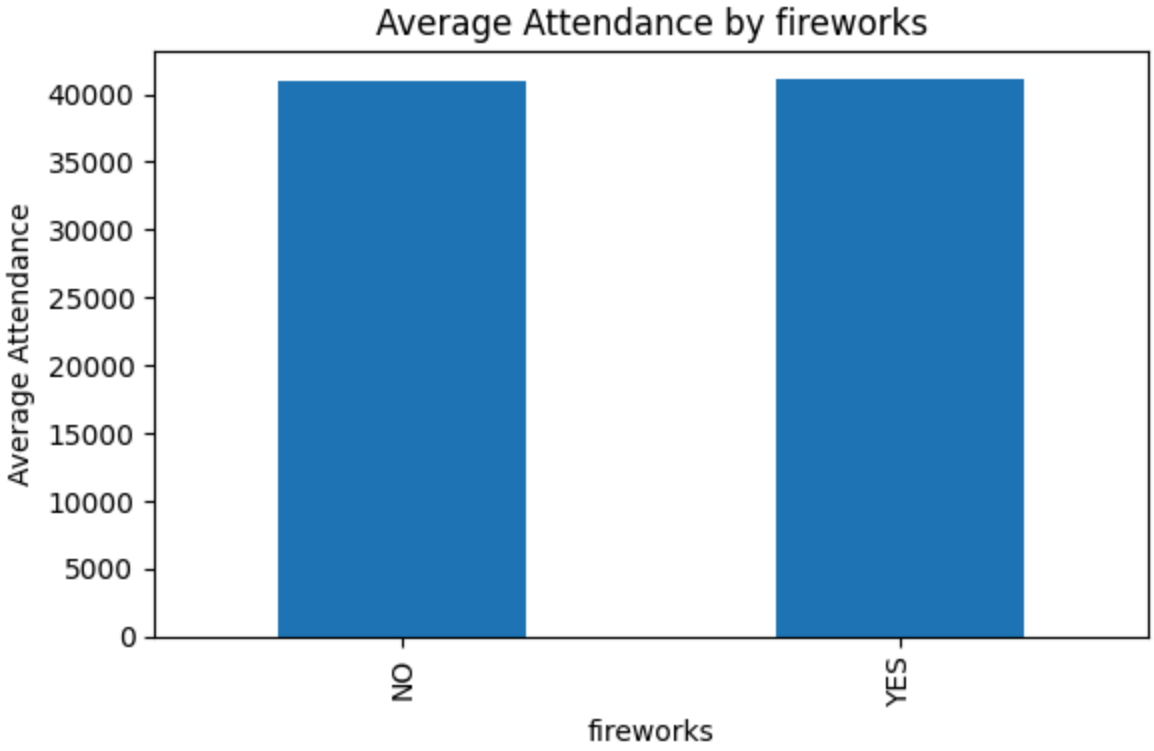
for col in columns_to_check:
    avg_attend = df.groupby(col)['attend'].mean()

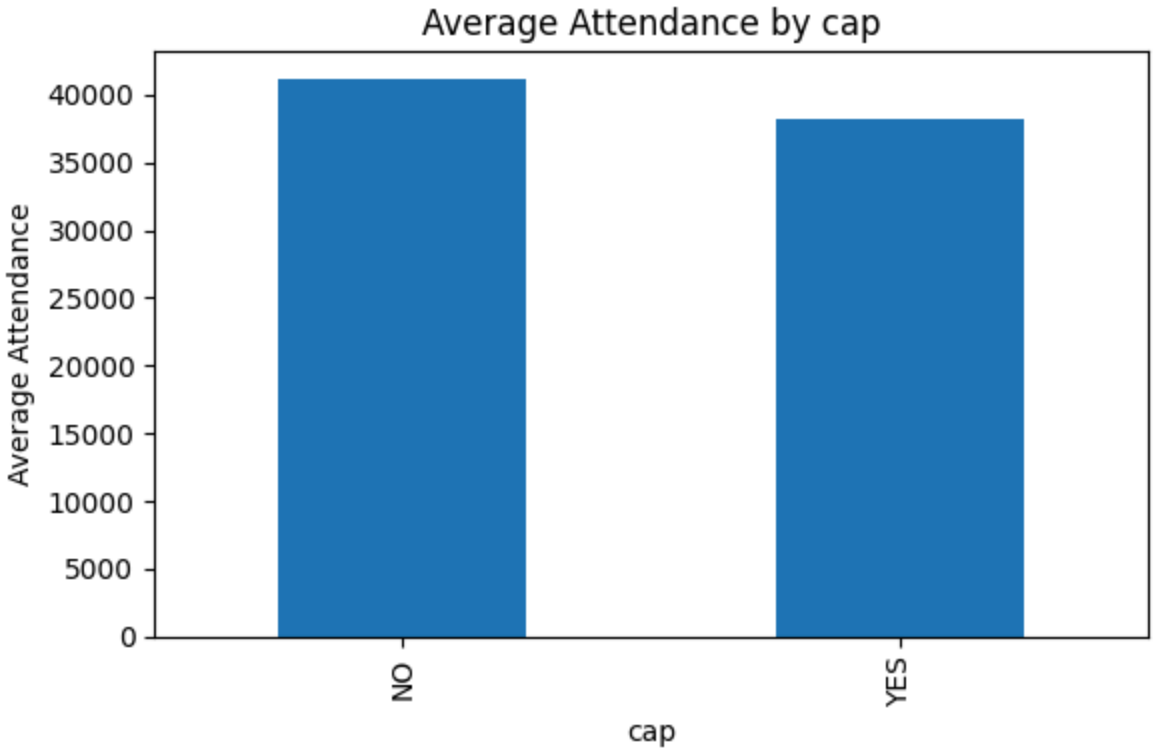
    # Plot bar chart
    plt.figure(figsize=(6,4))
    avg_attend.plot(kind='bar')

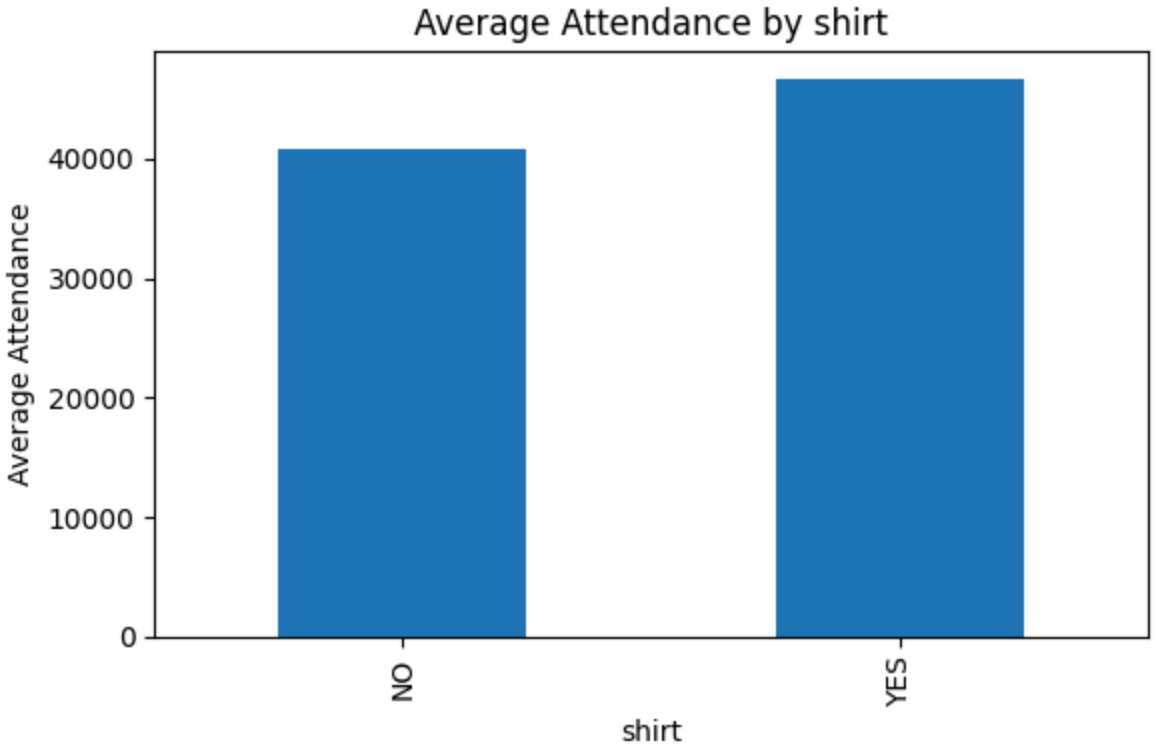
    plt.title(f"Average Attendance by {col}")
    plt.xlabel(col)
    plt.ylabel("Average Attendance")
    plt.tight_layout()
    plt.show()
```

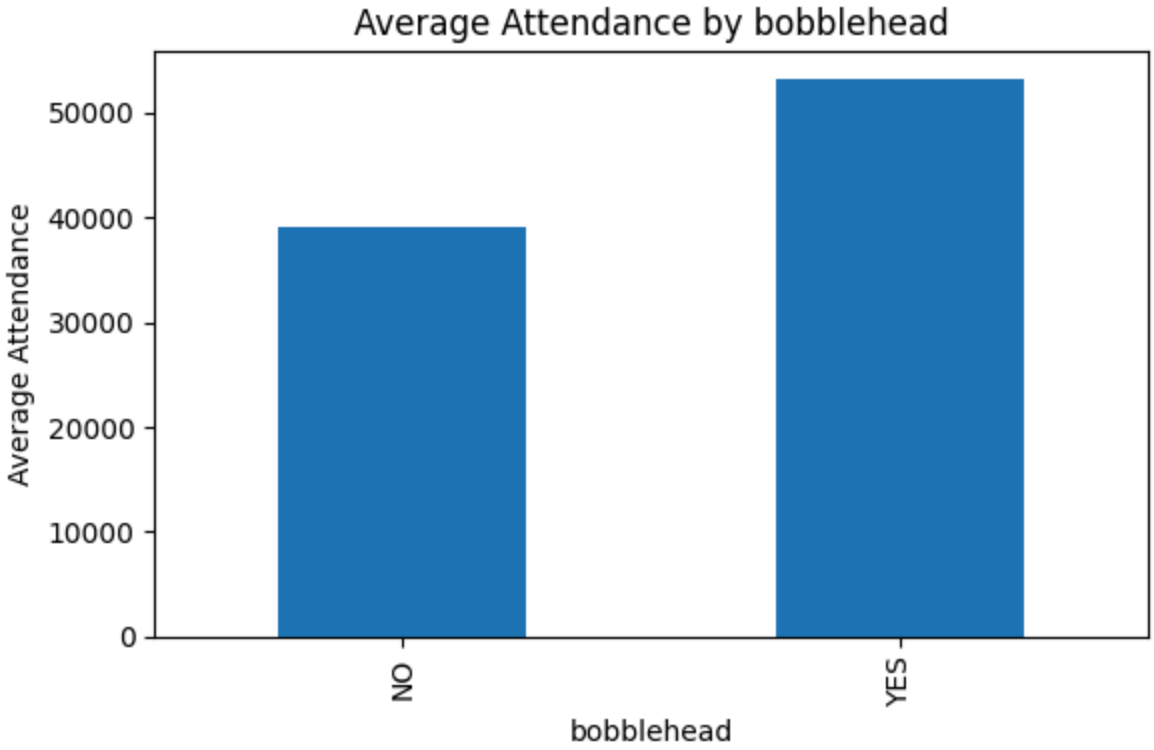

Average Attendance by skies
Average Attendance
40000
35000
30000
25000
20000
15000
10000
5000
0
Clear
Cloudy
skies

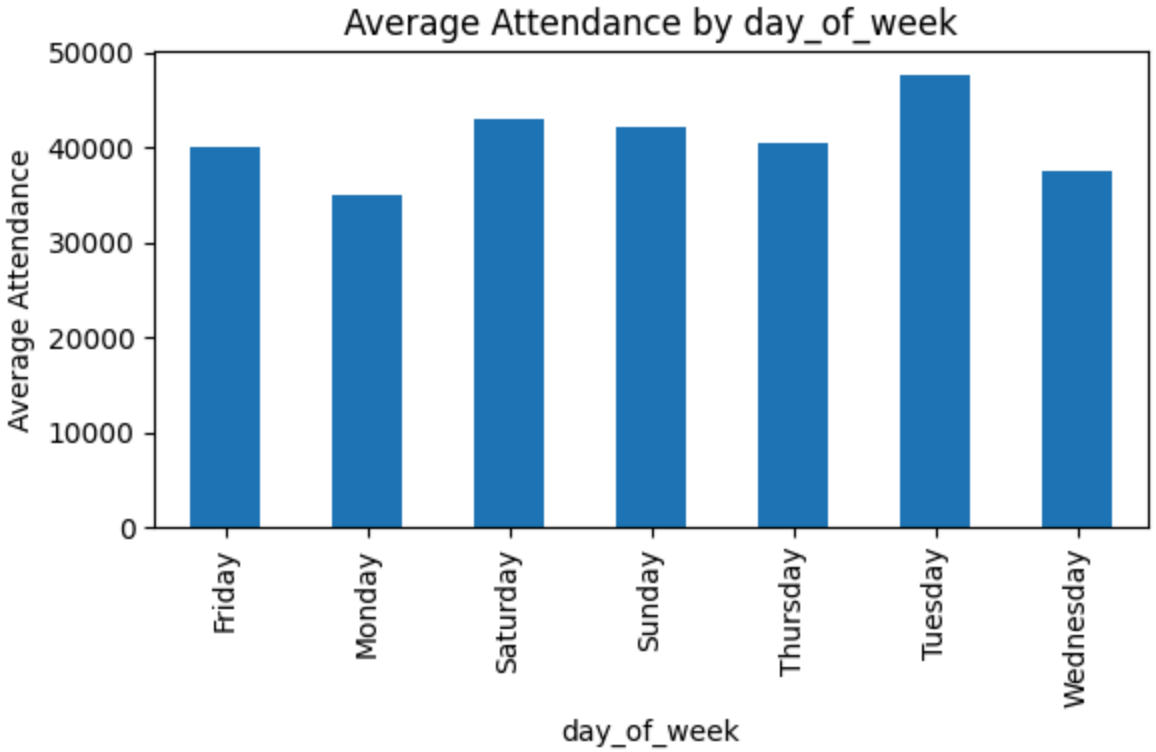

Average Attendance by day_night
Average Attendance
40000
35000
30000
25000
20000
15000
10000
5000
0
Day
Night
day_night

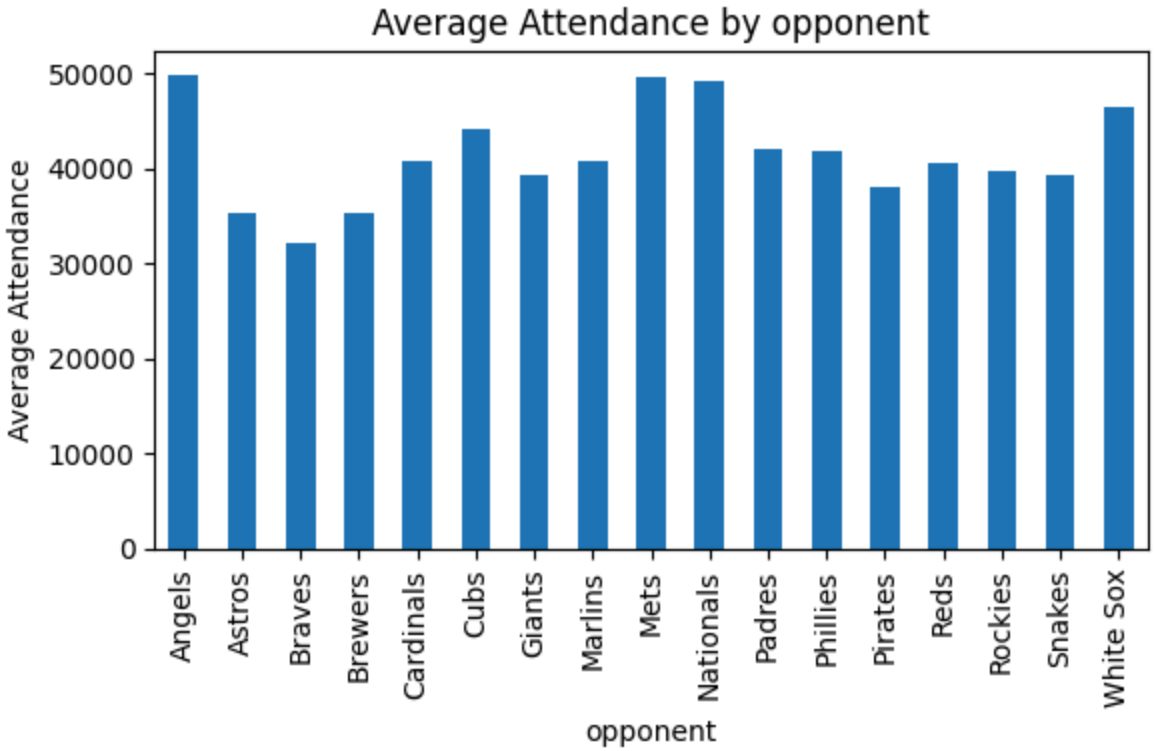

Average Attendance by fireworks
Average Attendance
40000
35000
30000
25000
20000
15000
10000
5000
0
NO
YES
fireworks

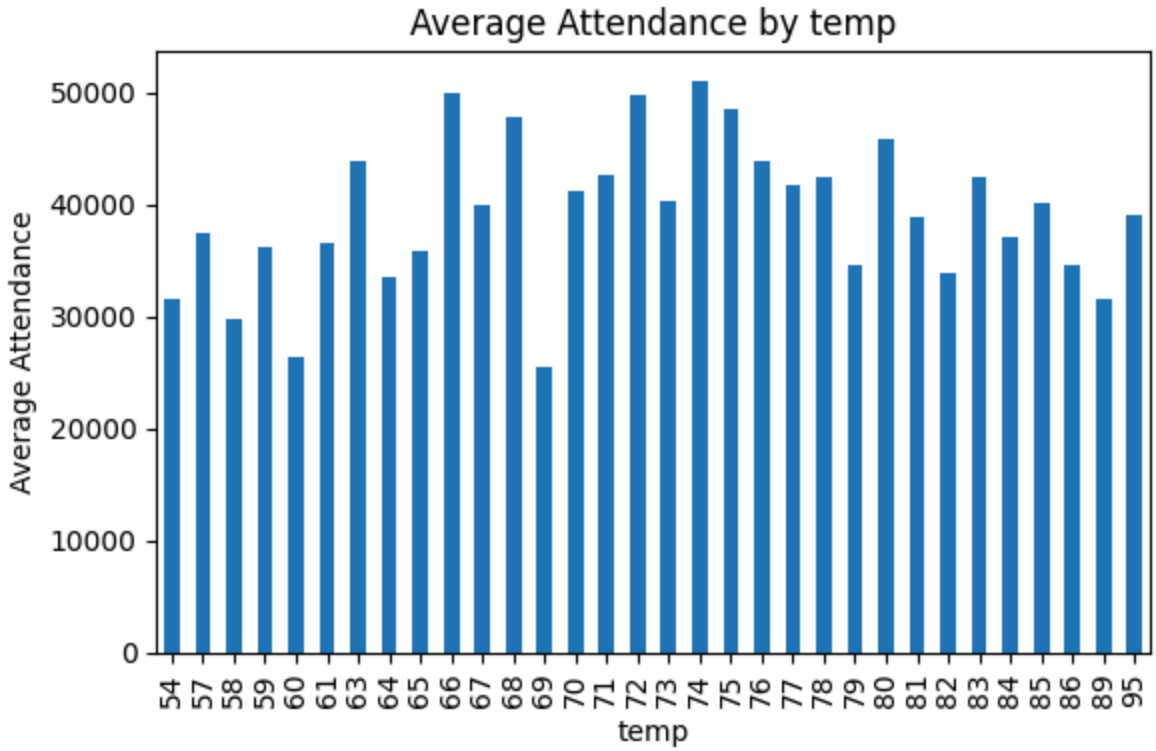

Average Attendance by cap
Average Attendance
40000
35000
30000
25000
20000
15000
10000
5000
0
NO
YES
cap

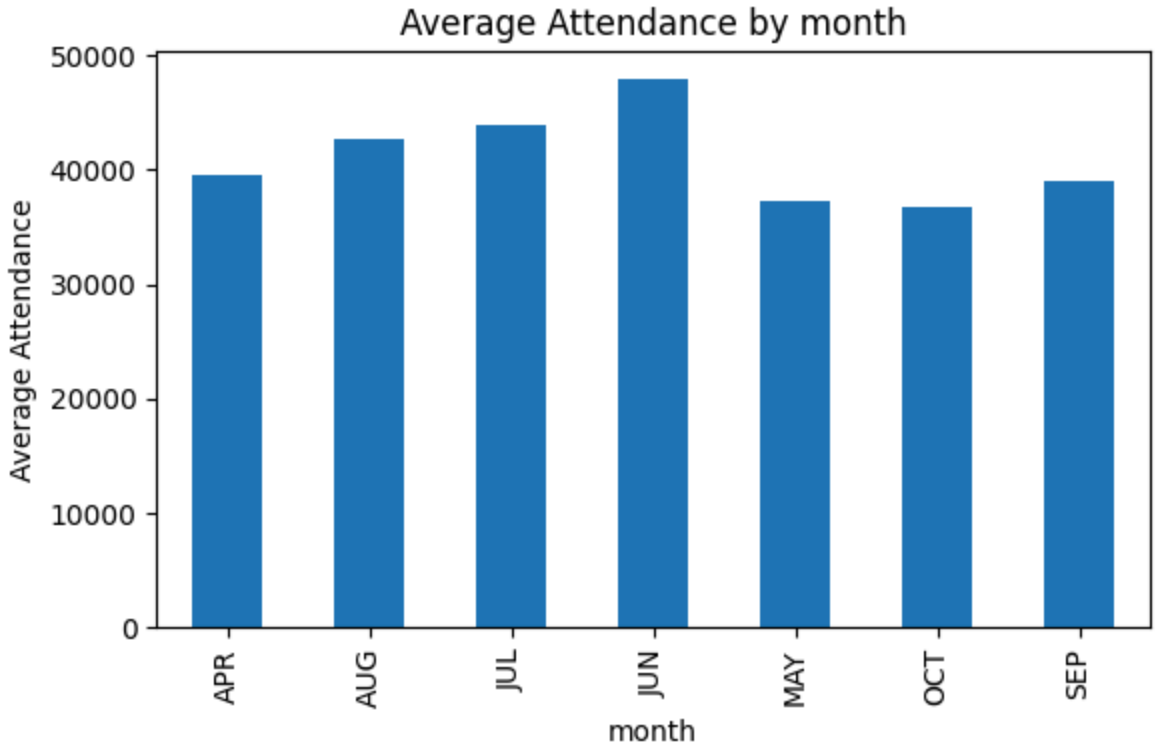

Average Attendance by shirt
Average Attendance
0
10000
20000
30000
40000
NO
YES
shirt

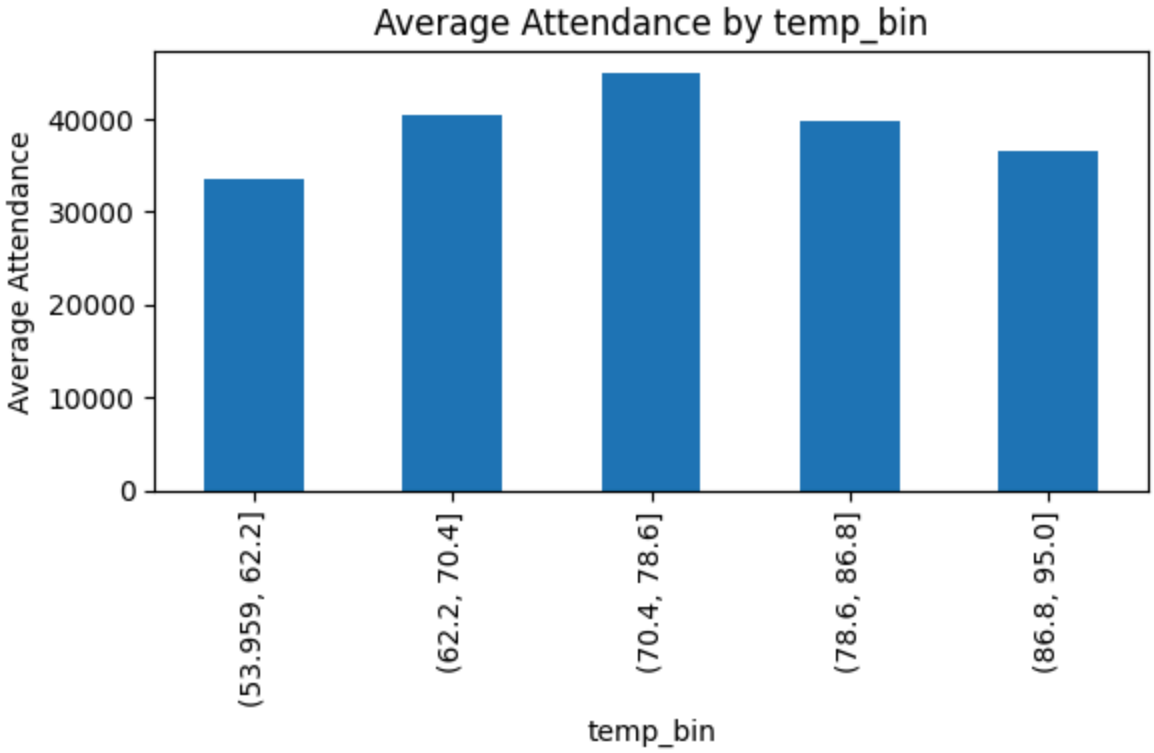

Average Attendance by bobblehead
Average Attendance
0
10000
20000
30000
40000
50000
NO
YES
bobblehead


Average Attendance by day_of_week
Average Attendance
50000
40000
30000
20000
10000
0
Friday
Monday
Saturday
Sunday
Thursday
Tuesday
Wednesday
day_of_week


Average Attendance by opponent
Average Attendance
50000
40000
30000
20000
10000
0
Angels
Astros
Braves
Brewers
Cardinals
Cubs
Giants
Marlins
Mets
Nationals
Padres
Phillies
Pirates
Reds
Rockies
Snakes
White Sox
opponent


Average Attendance by temp
Average Attendance
0
10000
20000
30000
40000
50000
54
57
58
59
60
61
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
89
95
temp


Average Attendance by month
Average Attendance
0
10000
20000
30000
40000
50000
APR
AUG
JUL
JUN
MAY
OCT
SEP
month



Task: We'll perform ANOVA tests to determine if there are statistically significant differences in mean attendance based on these factors, but it's pretty clear at this state which factors have the most impact.

In [19]:
```
# Run ANOVA for each categorical column

results = {}
for col in columns_to_check:
    # get unique non-null group values in a stable order
    groups_vals = df[col].dropna().unique()
    # build groups and drop groups with <=1 observation
    groups = [df.loc[df[col] == val, 'attend'].values for val in groups_vals
    groups = [g for g in groups if g.size > 1]

    F, p = f_oneway(*groups)
    results[col] = {'F': F, 'p': p, 'n_groups': len(groups)}


# Nicely print results
for col, res in results.items():
    print(f"{col} → F = {res['F']:.5f}, p = {res['p']:.5f} (groups={res['n_g
```

```
skies → F = 1.84240, p = 0.17854 (groups=2)
day_night → F = 0.15007, p = 0.69951 (groups=2)
fireworks → F = 0.00035, p = 0.98519 (groups=2)
cap → F = 0.23972, p = 0.62576 (groups=2)
shirt → F = 1.42845, p = 0.23559 (groups=2)
bobblehead → F = 40.44392, p = 0.00000 (groups=2)
day_of_week → F = 3.64403, p = 0.00319 (groups=7)
opponent → F = 1.37598, p = 0.18254 (groups=17)
temp → F = 1.28334, p = 0.23723 (groups=20)
month → F = 2.56720, p = 0.02585 (groups=7)
temp_bin → F = 3.67202, p = 0.00868 (groups=5)
```

Conclusion

Most factors show no significant difference in mean attendance ($p > 0.05$). Statistically significant effects are:

- bobblehead — large positive effect on attendance ($F = 40.444$, $p < 0.001$).
- day_of_week — attendance varies by weekday ($F = 3.644$, $p = 0.003$).
- month — attendance varies by month ($F = 2.567$, $p = 0.026$).
- temp_bin — attendance varies by temperature bin ($F = 3.67202$, $p = 0.00868$).

Binning temperature reduced p from 0.23 to ~0.0087, indicating a nonlinear effect: moderate temperatures increase attendance while extreme cold/hot reduce it.

In [20]:
```
# Check the occurrences of each opponent
print(df['opponent'].value_counts())
```

```
opponent
Giants       9
Rockies      9
Snakes       9
Padres       9
Cardinals    7
Mets         4
Brewers      4
Pirates      3
Cubs         3
Phillies     3
Reds         3
Astros       3
White Sox    3
Angels       3
Nationals    3
Braves       3
Marlins      3
Name: count, dtype: int64
```

Attendance for games is impacted by several factors. By gauging all possible factors, such as weather, promotional events and schedule, we can find variances to increase attendance.

Over the 2022 season, our attendance fluctuated between lows near 24,000 and highs of 56,000. These extremes are highly unlikely as they represent single data points. For example, opening day had high numbers, but we can't expect to repeat these data points each time.

We have little control over the weather or whether the game is played at night. Fortunately, these factors play a smaller role in attendance than other factors.

Promotional events such as giveaways and fireworks are within our control. Fireworks, shirts, and caps play little role in attendance. However, bobblehead giveaways have a significant impact on attendance.

For the schedule, Monday and Wednesday have lower attendance. Tuesday has the largest average attendance. Opponents also play a larger role in attendance, with the Angels, Mets, Nationals, and White Sox drawing the largest crowds. It is worth noting that we can't only play high attendance teams. Temperatures cannot be controlled but middle temperatures above 62 and below 86 draw higher attendance, as hot and cold temperatures deter attendance.

Overall guidance:

- April, August, July, and June have the highest attendance. Scheduling more home games in these months could boost attendance.
- May, October, and September have the lowest attendance. Either aim to improve these months with promotions or prepare for lower attendance.
- Bobblehead giveaways significantly boost attendance. Consider scheduling more bobblehead giveaway games.
- While they don't need to be completely cancelled, shirts, caps, and fireworks have little impact on attendance. Consider reducing these promotions to save costs.
- Attendance is higher between 62 and 86 degrees. Temperatures outside this range deter attendance. While we can't control the weather, we can consider scheduling fewer home games during extreme temperatures.